

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year EVEN Semester Examination 2024
COURSE NO: CSE 1203 COURSE TITLE: Object Oriented Programming
FULL MARKS: 60 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks and Course Outcomes (COs).
(iii) Use separate answer script for each section.

SECTION : A

Marks COs

- Q.1. (a) Explain the four fundamental principles of Object Oriented Programming (OOP) with simple examples. 05 CO1
(b) Design a class Student with the following data members and functions 05 CO1
i) Data Members: id, name and gpa
ii) Functions: display() and isoutstanding()
The isoutstanding() function returns true if $\text{gpa} \geq 3.75$. Write the complete class definition and a main() function that create two objects using constructor and check their outstanding status.

- Q.2. (a) Explain this pointer using an example. 02 CO1
(b) According to the statements in main() and output, define the class Circle with necessary codes. Assume radius as a private data member of the class. Here the method Area() calculates and prints the area of a circle. 04 CO1

<pre>void main(){ Circle c(5); Area(c); }</pre>	<pre>class Circle{ //write your codes };</pre>
---	--

- (c) Suppose a remote controller of an air cooler has four buttons for power on/off and controlling the temperatures up/down. The following table shows the functions of each button of the remote controller. Now using the concept of operator overloading implement the functionalities of the four buttons of the remote controller by writing a class Remote with necessary codes. Also write the required codes in main(): 04 CO3

Button	Function
+	Power On
-	Power Off
++	Increase the temperature by 1
--	Decrease the temperature by 1

- Q.3. (a) When do you use const keyword for a method in C++ and why? 02 CO1
(b) Add statements in the following classes A, B and C (if required) to implement the codes written in main(). Here method Avg() calculates and prints the average of a, b and c. Do not change the type of inheritance for each class. 04 CO2

<pre>class A{ private: int a; protected: int b; public: int c; };</pre>	<pre>class B:private A{ }; class C:public B{ };</pre>	<pre>void main(){ C cx(3,4,6); cx.Avg(); }</pre>
---	---	--

- (c) In OOP, it is often required to exchange information between two classes. Suppose there are two classes Doctor and Patient stores the information of doctors and patients respectively. Now write a C++ program to assign a doctor for each patient by the concept of aggregation. Also write codes in main() to implement it. 04 CO4

- Q.4. (a) Differentiate between i) static binding and ii) dynamic binding in C++. 02 CO3
(b) For the following classes and inheritance, write necessary codes in main() to obtain run-time polymorphism. Also update codes in classes if necessary. 04 CO3

<pre>class Doctor{ public: void Show(){ cout<<"Doctor"; } };</pre>	<pre>class Medicine:public Doctor{ public: void Show(){ cout<<"Medicine"; } };</pre>
--	--

- (c) Write a template class Test with necessary template method(s) to add i) two floating-point numbers and ii) two integer numbers using C++. Also write codes in main() to check the functionalities of the class. 04 CO3

SECTION : B

- Q.5. (a) Explain the difference between i) error and ii) exception. 02 CO1
 (b) Write a try-catch block when the program produces the following throws 04 CO4
 i) throws(1) prints "Integer Exception"
 ii) throws('a') prints "Char Exception"
 iii) prints "Unknown Exception" when above throws does not matched.
 (c) Suppose there are two Java classes A and B contains the methods ShowA() and ShowB() respectively. Now write codes in A and B to run these methods ShowA() and ShowB() simultaneously. Also write necessary codes in main() method. 04 CO4
- Q.6. (a) How does Java achieved multiple inheritance using interface? Explain with an example. 02 CO4
 (b) Write a Java program in which class SmartDevice implements two interfaces: 04 CO4
 – Camera with a method clickPhoto()
 – MediaPlayer with a method playMusic()
 Demonstrate calling both methods from the main() method.
 (c) Explain briefly 04 CO4
 i) how super keyword helps in method overriding
 ii) the role of ActionListener in handling swing events
- Q.7. (a) Write a Java swing program with the following specifications: 05 CO4
 i) Create a JFrame containing
 • Two JTextField for entering numbers
 • A JButton labeled "Add Numbers"
 • A JLabel to display the result
 ii) Implement ActionListener so that when the button is clicked:
 • the numbers from JTextField are added
 • The sum is displayed in JLabel
 (b) Draw a UML class diagram for the following scenario: 05 CO4
 • Vehicle is the base class with attributes: speed (int) and color (string). It has a method showDetails().
 • Car is derived from Vehicle and adds brand (string) as an attribute. It overrides showDetails() but also calls super.showDetails()
 • Bike is another sub class of Vehicle
 Indicate inheritance, override and attributes in the UML diagram.
- Q.8. (a) Consider the following scenario: 08 CO4
 A university system has:
 – Department which contains multiple Professors
 – Professor may belong to only one Department but a Department can exist even if no Professor is assigned yet.
 – Course objects are always taught by a Professor and a Course cannot exist without a Professor.
 Tasks:
 i) Draw a UML class diagram showing Department, Professor and Course.,
 ii) Indicate aggregation and composition relationships clearly.
 iii) Label multiplicities (e.g. 1..*, 0..1) for each relationship.
 iv) Explain in brief, why Department-Professor is aggregation and Professor-Course is composition.
 (b) Explain why singleton is useful in scenarios like logging or configuration management? 02 CO4

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2023

COURSE NO: CSE1203 COURSE TITLE: Object Oriented Programming

FULL MARKS: 60

TIME: 3HRS

N.B. (i) Answer any SIX questions taking any THREE from each section.

(ii) Figures in the right margin indicate full marks.

(iii) Use separate answer script for each section.

SECTION: A

		Marks	CO
Q.1.	(a) Discuss the use of public, private and protected access specifiers and their visibility in the class.	3	CO1
	(b) Imagine a tollbooth at a bridge. Car passing by the booth are expected to pay a 50 cent toll. Mostly they do, but sometimes a car goes by without paying. The tollbooth keeps track of the number of cars that have gone by, and of the total amount of money collected. Model this tollbooth with a class called tollbooth. The two data items are a type unsigned int to hold the total number of cars and a type double to hold the total amount of money collected. A constructor initializes both of these to 0. A member function called payingCar() increments the car total and adds 0.50 to the cash total. Another function, called nopayCar(), increments the car total but adds nothing to the cash total. Finally, a member function called display() displays the two total. Make appropriate member function const. Include a program to test this class. This program should allow the user to push one key to count a paying car, and another to count a nonpaying car. Pushing the Esc key should cause the program to printout the total car and total cash and then exit.	7	CO1
Q.2.	(a) Define class templates. How are they created? Create a template for bubble sort function.	4	CO4
	(b) Based on the following statements written inside the main() function, write necessary code in C++ in class 'Apple'. Assume that 'Apple' has a private data member 'size'.		
	<pre> class Apple{ String size; // Write your codes }; int main(){ Apple ax; Apple ay("Big"); Apple az(&ay); return 0; } </pre>	4	CO1
	(c) Explain the use of friend function with an example.	2	CO2
Q.3.	(a) When do you use const keyword before a method in C++ and why?	2	CO4
	(b) Add statements in the following classes A, B and C (if required) to implement the content set written in main() function. Here method Sum() prints the sum of data members a, b, and c. Do not change the type of inheritances for class B and C.		
	<pre> class A{ private: int a; protected: int b; public: int c; }; class B: private A{ }; class C: public B{ }; int main(){ C cx(10, 20, 30); cx.Sum(); return 0; } </pre>	4	CO2
	(c) Suppose a temperature controller has two buttons '+' and '-' to increase and decrease the temperature respectively. Now to implement the functionalities of the remote controller, write a 'Temperature' class that implements the statements given inside the main() function.		
	<pre> class Temperature{ int temp; // Write codes }; int main(){ Temperature t(20); ++t; // increase by 1 --t // decrease by 1 return 0; } </pre>	4	CO3
Q.4.	(a) What is the usefulness of pure virtual function in C++?	2	CO3
	(b) For the following classes and inheritance, write necessary codes in main() to obtain run-time polymorphism. Also update codes in classes if required.		
	<pre> class Player{ public: void Show(){ cout<<"I am player"; } }; class Cricket: public Player{ public: void Show(){ cout<<"I am Cricket Player"; } }; int main(){ //Write codes return 0; } </pre>	4	CO2 CO3
	(c) Suppose in a Java program class AX and BX contains the methods DisplayAX() and DisplayBX() respectively. Now write codes in AX, BX and main() to run	4	CO4

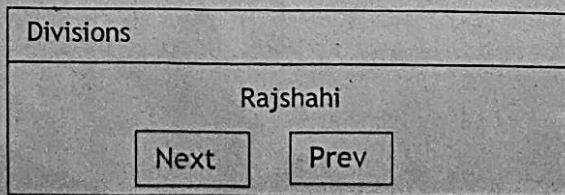
these two methods simultaneously using multithreading concept of Java.

SECTION : B

- Q.5. (a) Explain why UML is important in writing a program? 2 CO4
- (b) Draw an UML diagram for the following C++ classes and relation-
class A{ class B: Public A{
private: int ax; public: void Display(){
protected: float ay; cout<<ay<<endl;
public: void SetA(int x, float y){ }
ax=x; ay=y; };
} };
- (c) In OOP, sometimes it is required to exchange information between classes. Suppose in the following 'Dept' and 'Student' classes stores the department and student information respectively of an institution. Here 'Student' class requires information from 'Dept' class. Now write a program in C++ that implements aggregation in 'Student' class using 'Dept' object and then explain the concept of aggregation. Also write the statements in main() to implement it. 3 CO2
- ```

class Dept{ class Student{ int main(){
 String dname; String sname; // Write codes here
 //Write codes here Dept *dn;
}; // Write codes here }
}; };

```
- Q.6. (a) Differentiate among JRE, JDK and JVM. 3 CO4
- (b) Write a method MyArray() in Java that returns an array {10, 40, 50, 70} anonymously. Also write the corresponding statements in main() to call it. Finally display the array using toString() method. 3 CO4
- (c) Write statements in Java to call Display() and Show() in main() function-  
public class MyFirst{  
public static void main(String [] args){  
//Write codes here  
}  
private void Show(){  
System.out.println("Show"); }  
Private static void Display(){  
System.out.println("Display");  
} } 4 CO4
- Q.7. (a) Write two interfaces I<sub>1</sub> and I<sub>2</sub> that contain methods called "methodA" and "methodB" respectively. Now, write a class "interfaceTest" and implement the interface B in such a way that both methods "methodA" and "methodB" are implemented in the "InterfaceTest" class. 4 CO3
- (b) The Singleton pattern is used in situations where a single instance of a class needs to coordinate actions across the system, such as in a logging or configuration system. It ensures that there is only one instance of the class, which can be accessed globally, preventing unnecessary duplication and ensuring consistency across the system. Now design a Singleton class "MySingleton" in Java and write necessary statements in main() to create one object. 4 CO1 CO4
- (c) Define Multithreading. How is a thread created in Java? 2 CO4
- Q.8. (a) Differentiate between abstract class and interface. Explain what will happen if we want to change the value of a field in interface after initialization. 4 CO3
- (b) Design and write a Java program using Swing library to implement the following frame titled as "Divisions". The program displays the divisions of Bangladesh namely as "Barishal", "Chattogram", "Dhaka", "Khulna", "Rajshahi", "Rangpur", "Mymensingh", and "Sylhet". When clicks the "Next" button, next division is displayed and vice-versa it displays previous division when clicks "Prev" button. Initially "Rajshahi" division is to be displayed. 6 CO2 CO3 CO4



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**RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**1<sup>st</sup> Year Even Semester Examination 2022**

**COURSE NO: CSE 1203**

**COURSE TITLE: Object Oriented Programming**

**FULL MARKS: 60**

**TIME: 3 Hours**

- N.B.** (i) Answer any **SIX** questions taking any **THREE** from each section.  
(ii) Figures in the right margin indicate full marks.  
(iii) Use separate answer script for each section.  
(iv) COs- Course Outcomes

**SECTION A**

**COs Marks**

- Q.1. (a) Explain how data encapsulation and data abstraction are obtained by using C++ programming with an example. CO1 02
- (b) John Smith is a nice person working in an IT company. He has an official car for his own use although his family members are eager to use it. But Mr. Smith is very strict to let them use his car. However Mr. Smith has bought a bike for his family members. John Smith is a health conscious person and he hits the gym every weekend to stay fit and rejuvenate. He dedicated his free time to social work, tirelessly helping those in need within his community. Now from above scenario, guess and identify the object oriented feature in it. CO1 03
- (c) Dengue (DEN-gee) fever is a tropical disease by virus carried by mosquitoes. The virus can cause fever, headaches and pain throughout the body. People with symptoms of dengue fever or recent exposure to the disease undergo the Dengue IgG test in the hospital. The following table shows the IgG values and their corresponding results: CO1 05

| IgG Value                         | Result    |
|-----------------------------------|-----------|
| 1.64 or less                      | Negative  |
| more than 1.64 but less than 2.85 | Equivocal |
| 2.88 or greater                   | Positive  |

To automate the process of patient management in the hospital, define a class Dengue { } and write methods to do the followings:

- (i) Register each patient with ID and Name (ii) Set IgG test value to a patient (iii) Assign a doctor for a patient (iv) Display ID and Name of a patient with Dengue Positive (v) Display the IgG status for a particular patient and (vi) Count no. of Dengue negative, Equivocal and positive patient in the hospital.

- Q.2. (a) What is the usefulness of Constructor in C++? Name the different types of Constructors used in C++. CO1 02
- (b) Find the errors (if any) in the following C++ program otherwise show the output. CO1 03

```
#include <iostream>
using namespace std;
class Person {
 String name;
public:
 Person (String s){
 name = s;
 }
 String getName () {
 return name;
 }
};
void main () {
 Person q ();
 Person p ("John");
 Cout << p.getName ();
}
```

- (c) Suppose a class Circle has a private data member 'r' and now do the following if C1 and C2 are the objects of circle class CO1 05
- (i) Write Constructor to initialize C1 with r = 5 (ii) Write a Constructor to copy the value of r of C1 to C2 (iii) Write a method getArea () that returns the area of a Circle. Note that radius cannot be changed here and (iv) Write the statements required in main () to execute (i) – (iii).

- Q.3. (a) Draw different types of inheritance using class diagram. CO2 02
- (b) Let a, b, c are the inputs from keyboard denote the angles of a triangle. There are three classes A, B and C have a protected data members x, y and z respectively corresponds to a, b and c. The CheckTriangle () method checks whether the above three angles form a triangle or not. Now write the required Constructors/methods to execute the statements given in main () function shown below CO2 03

```
#include <iostream>
using namespace std;
Class A {
 protected: int x;
};
Class B: protected A {
 protected: int y;
};
Class C: private B {
 protected: int z;
};
void main () {
 Cin >> a >> b >> c;
 C cx (a, b, c);
 cx.ChecTriangle ();
}
```

- (c) There are two classes Person and Student written in C++ shown below. The data members of each class are initialized by its Constructor only and each class has a Display () method to display own class data members only. Now update the following code to print nid, name and dept of a student correctly. CO2 05

|                                                                                               |                                                                                        |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <pre> class Person {     int nid;     string name;     // write code here }; </pre>           | <pre> class Student: public Person {     string dept;     // write code here }; </pre> |
| <pre> int main (){     Student s(100, "John", "CSE");     s.Display ();     return 0;} </pre> |                                                                                        |

- Q.4. (a) List some significant of static data and static member function in C++. Develop a program to demonstrate static keyword in C++. CO3 03  
(b) Discuss the role of friend function in C++. CO3 03  
(c) Describe how function overriding is different from function overloading? Design a C++ program involving overriding of member function. CO4 04

#### SECTION B

- Q.5. (a) Implement Binary+ operator using operator function in C++. CO3 04  
(b) Overload the following function:  
void sum (int a, int b);  
int sum (int a, int b, int c);  
double sum (double a, double b);  
double sum (double a, int b);  
(c) Explain the use of super keyword in Java. CO3 02  
Q.6. (a) Write added/updated codes only to complete the following three tasks. CO4 05

|                                                                                                                 |                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> class InvalidUserException { Extend the exception class} {     // write your answer for (i) here } </pre> | <pre> class ExceptionDemo {     void UserCheck (int age, int work_experience) throws InvalidException {         // Write your answer for (ii) here     } } </pre> |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

public static void main (String args[]){
 Exceptiondemo obj = new Exceptiondemo ();
 Scanner input = new Scanner (System.in);
 int age = input. nextInt ();
 // write your answer for (iii) here
 obj.usercheck (age, work_experience);
}

```

- (i) Create a user-defined exception class named InvalidUserException that extends the Exception class and points the message provided below using the super constructor.  
(ii) Provide logical implementation so that if age is below 50, then throw an exception of InvalidUserException class with the message "Ineligible for Elderly pension because of age", if work\_experience is below 20, then throw an exception of InvalidUserException class with message "Ineligible for Elderly pension because of work experience."  
(iii) Invoke the UserCheck () method using proper try-catch block. Your catch-block should point a message "Caught exception and print the correct exception" and print the correct message from (ii).  
(b) Explain Generic functions and generic class. Write C++ program to find sum of three integer number using template class. CO4 05  
Q.7. (a) The Aggregation and Composition are two important relationships in Unified Modeling Language (UML). Then how do you implement (i) aggregation and (ii) composition using C++. CO4 02  
(b) Draw an UML diagram for the following classes and relation  
class A {  
private: int x;  
protected: double y;  
public: char z;  
void getData (int a, double b, char c){  
x = a;  
y = b;  
z = c;}  
private double Sum (){  
return (x + y);}  
class B: public A {  
private: int x;  
public void getData (int a){  
x = a;}  
(c) Write a Java program that implements 100% abstraction with an example. How default method can be written inside an interface in Java and how can it be called from main () method. CO4 05

- Q.8. (a) Write the code to get a java GUI application like image below that has the functionality of converting foot into inch after pressing the convert button. Assume, all packages are imported. User can input decimal numbers in the input field (i, e 10, 5.5 etc). Formula: 1 foot = 12 inch. CO4 05

|                                      |                                   |
|--------------------------------------|-----------------------------------|
| Convert - <input type="checkbox"/> X |                                   |
| Foot                                 | <input type="text" value="1"/>    |
| Convert to inch                      |                                   |
| inch                                 | <input type="text" value="12.0"/> |

- (b) Discuss about the use of virtual functions in the context of inheritance which impacts the behaviors and functionality of derived classes? Demonstrate an example. CO2 05

**RAJSHAH UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**1<sup>st</sup> Year Even Semester Examination 2021**

**COURSE NO: CSE 1203**

**COURSE TITLE: Object Oriented Programming**

**FULL MARKS: 60**

**TIME: 3 Hours**

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.  
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(iii) Use separate answer script for each section.

**SECTION : A**

COs    Marks

- Q.1. (a) Write down four important properties of Object Oriented Programming (OOP). CO1    2  
(b) Generally computer viruses are programmed to harm our computer by damaging programs, deleting files or reformatting the hard drive. The well known viruses are Trojan, Nimda, Conflicker, Mydream, Melissa etc. Each virus has some similar properties like date of publish, type of destruction and whether it is removable or not. Now from the above scenario guess the class and objects and the member of the class. CO1    3  
(c) Suppose ABC Global is an Internet Service Provider (ISP) company and you are working as a programmer there. The company provides internet connectivity to its customers and collects money from them. In each month, the company inputs monthly data usages and prepare a bill for each customer. To automate the process you need to do the following using C++ programming. Assume the cost for 1KB of data usage is Tk. 1.5. CO1    5  
i) Create a class "Customer" with its data members and methods.  
ii) Register each customer by customerID, name, address and mobile number.  
iii) Prepare monthly bill for a customer.  
iv) Calculate total earning of the company.

Sample Customer Bill:

```

**** BILL ****
Customer ID: 3
Name: xxxx
Data used: 5KB
Amount: Tk. 7.5
ABC Global Company

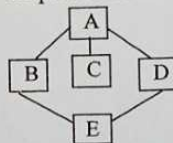
```

- Q.2. (a) How and why is the scope resolution operator (::) used in class definition. CO1    2  
(b) For the following Circle class write a copy constructor that copies the radius from another circle object. Also write how to invoke the copy constructor from main() function. CO1    3
- ```

class Circle{
    double radius;
public:
    Circle(double r){
        radius=3;
    }
    // Write your code here
};

int main(){
    // write your code here
    return 0;
}

```
- (c) Write a C++ program that defines a class "Rectangle" that takes length and width of rectangle from keyboard and then perform the following operations by writing methods in it. CO1 5
i) Declare length and width as private.
ii) A method to initialize length and width.
iii) A method to display length and width. No data member should be changed here.
iv) A method to display area of the rectangle.
v) A method to return the diagonal of the rectangle.
- Q.3. (a) Describe the benefits of inheritance in object oriented programming. CO2 2
(b) Define the classes A, B, C, D and E to implement the following inheritance class diagram- CO2 3



- (c) There are two classes "Person" and "Student" written in C++ are given as below. The data members of each class are initialized by its constructor only. And each class has a "Display()" function to display its data members only. Now update the following code so that it prints nid, name and dept of a student object correctly. CO2 5

<pre> class Person{ private: int nid; string name; // write your code here }; </pre>	<pre> class Student: public Person{ string dept; // write your code here }; </pre>	<pre> int main(){ student s(100, "John", "CSE"); s.Display(); return 0; } </pre>
--	--	--

- Q.4. (a) Show the importance of friend function and friend class with example. CO2 3
 (b) Consider the situation where class Book is inherited by two other classes fiction and nonfiction. Both these classes are inherited into another new class library. Then analyze the situation if any data/member function of class Book is accessed by an object of class library? Illustrate how to resolve this issue? CO2 4
 (c) Briefly tell the procedure of making a class abstract in C++ by giving an example. CO3 3

SECTION : B

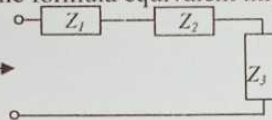
- Q.5. (a) Differentiate between, i) Function overloading, and ii) Function overriding. CO3 2
 (b) Based on the following main() function design the necessary class and method to return the sum of a set of numbers: CO3 3

```
int main(){
    A a;
    cout<<a.Sum(3,4)<<endl;
    cout<<a.Sum(5,4,6)<<endl;
    cout<<a.Sum(5,4,6)<<endl;
    return 0;
}
```

- (c) In AC circuit impedance is very important component consists of real and imaginary values. To calculate the equivalent impedance of the following AC circuit where three impedances $Z_1=3+j4$, $Z_2=5-j2$ and $Z_3=j6$ are connected in series. Then write a C++ program that implements the operator overload concept and prints the output as $8+j8$. Use class circuit and initialize the impedance values using two data members real and img by a constructor. The formula equivalent impedance is $Z=Z_1+Z_2+Z_3$. CO3 5

Equivalent impedance

$$Z=Z_1+Z_2+Z_3$$



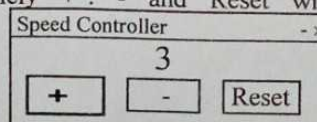
- Q.6. (a) Explain the way of creating a Template class and template function with example. CO4 4
 (b) Explain the statement: `vector<int> ivec(4,3)` CO4 2
 (c) Can a constructor throw an exception? Summarize the way to handle error when the constructor fails? CO4 4
 Q.7. (a) Give three advantages and disadvantages of immutable class. CO4 4
 (b) A programmer has created an AssetLocation class to represent the location of company assets. Some assets are mobile and their location must be updated regularly (e.g., vehicles); others are static and will never be updated (e.g., warehouses); The class contains a string describing the assets; an int recording a unique identifier and two values to represent valid latitude ($-90^\circ \leq \phi \leq 90^\circ$) and longitude ($-180^\circ \leq \theta \leq 180^\circ$) values respectively. All fields are initially mutable and set by the constructor. Write Java code that implement AssetLocation as described. CO4 4
 (c) The following source code shows compile time error. Identify the statement that causing the error and explain the reason behind it. Give solution to overcome this. CO4 3

```
class Toyota{
    // class Toyota members
}
class Honda
{ // class Honda memnbers}
class Mitsubishi extends Toyota, Honda
{ // class Mitsubishi members
}
```

- Q.8. (a) The aggregation and composition are two important relationships in Unified Modeling Language (UML). Then how can you implement-i) Aggregation, ii) Composition using C++ Program. CO4 2
 (b) Draw an UML diagram for the following classes and relation- CO4 3

<pre>class A{ private: int x; protected: double y; public: char z; void getData(int a, double b, char c){ x=a; y=b; z=c; } double Sum(){ return x+y; } }</pre>	<pre>class B: public A{ private: int x; public: void getData(int a){ x=a; } };</pre>
--	--

- (c) Write a program in Java to design and implement the following window frame named as "speed controller". There are three buttons namely "+", "-", and "Reset" with the following functionalities- CO4 5
 i) when clicks "+" button label increases by 1,
 ii) when clicks "-" button label decreases by 1,
 iii) when clicks "Reset" button label resets to 0.



Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2019
COURSE NO: CSE 1203 COURSE TITLE: Object Oriented Programming
FULL MARKS: 72 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

SECTION : A

Marks

Q.1. (a) What is object oriented programming? Briefly explain the object oriented features provided by C++. 4

(b) Consider the following class

```
class student {
    private : int *roll;
             float *cgpa;
}
```

Now:

(i) What is constructor? Write appropriate parameterized constructor for the above class. 3

(ii) What is copy constructor? Write a copy constructor for the above class. 3

(iii) What is destructor? Write necessary destructor for the above class. 2

Q.2. (a) Demonstrate inline function with necessary C++ code. Explain how inline function can improve execution speed. 4

(b) Write down the difference between reference and pointer. 2

(c) Consider the following class and given main function. 6

```
class student {
    private : string roll;
             float cgpa;
    public : student(string r, float cg) {
        roll=r;
        cgpa=cg;
    }
    Void printinfo () {
        cout<<roll<< " " <<cgpa<<endl;
    }
}
```

// write additional codes if necessary

};

int main(){

 student *arr[10];

// write additional codes if necessary

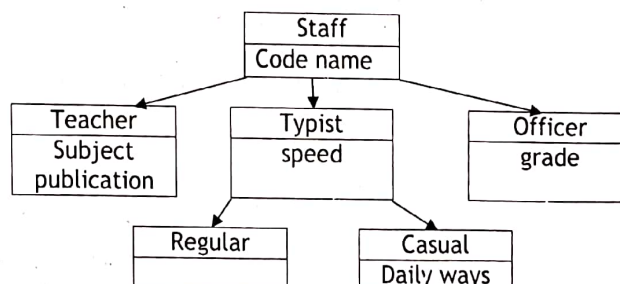
//-----

 return 0;

}

Now write necessary/additional C++ code so you can take ten (10) roll and cgpa as input from user and store them in the array arr[] given in the main function of the above code. Later sort this array in descending order and finally print the values of the sorted array arr[] using printinfo() function.

Q.3. (a) An educational institution wishes to maintain a data of its employees. The hierarchical relationships of related classes are as shown in below figure. Define all the classes to represent above hierarchy and define functions to retrieve individual information as and when required. 6



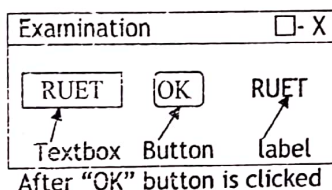
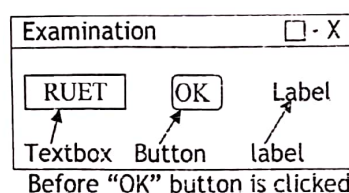
(b) What is operator overloading? State the rules for operator overloading. 4

(c) Define data abstraction with example. 2

- Q.4. (a) Answer the following: 4
- Define concept of virtual base class
 - Give reason for assigning protected visibility specifier to a class member.
- (b) State any four points of differentiation between compile time polymorphism and run time polymorphism. 4
- (c) Define a class named "Bank-Account" to represent the following members: 4
- Data members:
- Account Number
 - Name of Depositor
 - Account Type
 - Balance Amount
- Member functions
- Initialize members
 - Deposit amount
 - Withdraw amount
 - Display Balance
- Write a C++ program to test the Bank account class for 10 customers.

SECTION : B

- Q.5. (a) What are the differences between abstract class and interface? 4
- (b) Write a program in Java to illustrate the implementation of multiple inheritances through interfaces. 4
- (c) Write notes on: Packages, scanner classes and stream classes with example. 4
- Q.6. (a) What is abstraction in Java? With appropriate java code, explain how java interface can provide complete abstraction. 6
- (b) Why main method in Java is declared as public static? Explain briefly. 3
- (c) Demonstrate application of variable length argument with proper Java code. 3
- Q.7. (a) Consider the following Java class that is intended to represent a specific day in an eight-week University term. 9
- ```
public class TermDay {
 public int day; //The day of the week as a number 0-6
 public int week; //The week of the term as a number 0-7
};
```
- Create a class Encapsulated TermDay which applies the principles of data encapsulation as an alternative to TermDay. Your modified class should throw an exception if an invalid day of the week or week number is specified.
  - Create a class Immutable TermDay that is an immutable version of TermDay.
  - By applying one or more appropriate design patterns and adapting Immutable TermDay appropriately, show that how to ensure that only one Immutable TermDay object is ever created for a given day/week combination.
- (b) Explain method overriding with a suitable example program. 3
- Q.8. (a) Write down the difference among program, process and thread. 5
- (b) Write a Java code that has two threads where one thread is printing the output "RUET CSE" for 1000 times and the other thread is printing the output "18 SERIES" for 1000 times provided that "RUET CSE" is printed for 1000 times at first and only after that "18 SERIES" is printed for 1000 times. 4
- (c) Write necessary Java code in order to create the following graphical user interface while satisfying the conditions, 3
- Use flow layout
  - When "OK" button is pressed the input in the test field is shown in the label.
  - Define the default close operation for the interface.



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**RAJSHAH UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**1<sup>st</sup> Year Even Semester Examination 2018**

**COURSE NO: CSE 1203**

**COURSE TITLE: Object Oriented Programming**

**FULL MARKS: 70**

**TIME: 3 HRS**

- N.B.** (i) Answer any **SIX** questions taking any **THREE** from each section.  
(ii) Figures in the right margin indicate full marks.  
(iii) Use separate answer script for each section.

**SECTION : A**

- Q.1.** (a) What is object oriented programming? Explain the features of object oriented programming 3  
(b) Consider the class 6  

```
class A{
private: int x;
 int y;
}
```

Now (i) Write down the constructor and destructor of the above class. Also explain the role of constructor and destructor. (ii) Write down accessor and mutator for the above class members.  
(c) Explain the concept of "this" pointer in C++ and mention the cases where it is not usable. 3
- Q.2.**

```
#include<iostream>
using namespace std;
class ct2{
int *a;
public:
ct2(int x){a=new int; *a=x;}
ct2(const ct2 &o);
~ct2(){cout<<"Destructor"<<endl; delete a;}
void show(double y){cout<<y<<endl;}
void show(float y){cout<<y<<endl;}
};
ct2::ct2(const ct2 &o){
a= new int;
a=o.a;
}
int main(){
ct2 ob(4);
ct2 ob1=ob;
ob.show(10);
return 0;
}
```

Answer the following questions:  
(a) In the above program, which type of errors will be occurred? Find the error and solve it. 4  
(b) Write down the general form of copy constructor. Which statements would invoke the copy constructor of a class in general. 4  
(c) Add some codes to above program, where an ambiguity will be occurred base on default arguments plus overloading. 4
- Q.3.**

```
#include<iostream>
using namespace std;
class question03{
int a;
public:
question03(){ a=0;}
question03(int n){a=n;}
int get(){return a;}
question03 operator+(int n);
};

question03 question03 :: operator + (int n){
question03 temp;
temp.a=a+n;
return temp;
}
int main(){
question03 o1, o2(6);
o1=14+o2;
cout<<o1.get()<<endl;
return 0;
}
```

Answer the following questions:

- (a) In the above program, which statement is not correct? Why?  
 (b) Solve the above problem using friend operator function.  
 (c) Add some coes to the corrected program, where overload the [] subscript operator.

Q.4. (a) What is abstract class? How can you create an abstract class in C++? Demonstrate it with necessary code.

(b) Consider the following class:

```
class A{
protected: int x;
private: int a;
public: int m;
};
```

```
class B {
};
```

Now if there is a class B, then answer the following question with corresponding c++ code, When (i) class B inherits A as public, how can you access the given members of class A inside class B (ii) class B inherits A as protected, how can you access the members of class A inside class B (iii) Class B inherits A as private, how can you access the members of class A inside class B (iv) class B inherits A as public, how can you explicitly invoke constructor of A from class B.

#### SECTION : B

Q.5. (a) Why is Java considered as platform independent? Explain briefly.

(b) Read the following code carefully and then answer the following questions:

```
#include <iostream>
using namespace std;
class A{
private: int *x;
public:
A(){x=new int(0);}
A(int v){x=new int(v);}
void setx(int s){*x=s;}
void print(){cout<<"x"<<endl;}

~A(){
/*delete(x);*/
}
};

int main(){
A ob1(10);
A ob2(ob1);
ob1.print();
ob2.print();
ob1.setx(1000);
ob1.print();
ob2.print();

/*
A ob3;
ob3=100+ob1;
A ob4;
ob4=ob1+ob2;
ob3.print();
ob4.print();
*/

return 0;
}
```

- (i) Write down the output of the following code  
 (ii) What will happen if the commented section in the destructor is uncommented?  
 (iii) Identify and explain the problem in the given code, afterwards write the proper code that resolves the problem  
 (iv) Now consider that the commented section in the main() method has been uncommented, therefore write down the required code that needs to be incorporated in order to execute the main() method without any error.

Q.6. (a) Explain how can you prevent a class from being inherited?  
 (b) Read the following code carefully and answer the following questions:

```
#include<iostream>
using namespace std;
class A{
protected: int x;
public:
A(){x=0;}
A(int value){x=value;}
void print(){cout<<"Value of x="<<x<<endl;}
};
class B: public A{
public:
B():A(0){ }
B(int value): A(value){ }
void print(){cout<<"Value of x="<<x+100<<endl;}
};

int main(){
B ob1(10);
ob1.print();
A *ob_p;
ob_p=new B(10);
ob_p->print();
}
```

- (i) Write down the output of the above code  
 (ii) Identify function overloading and overriding in the above code and explain how they are different  
 (iii) In the above code object of class B has been created in the following two ways B ob1(10); and ob\_p=new B(100);  
 Now except the fact of ob\_p being a pointer of class A, what is the actual difference associated with these two type of object creation.

- (iv) Identify the problem in the above code. Why this problem has occurred and using which mechanism you can solve this problem? Explain briefly.

```

package ja03;
public class JA03{
 public static void main (String[] args){
 GeometricObject c_obj=new Circle(4.0);
 GeometricObject r_obj=new Rectangle(5.0,10.0);
 System.out.println(c_obj.getArea());
 System.out.println(r_obj.getArea());
 }
}

class GeometricObject{
 public double getArea(){return 0.0;}
}

class Circle extends GeometricObject{
 private double radius;
 Circle(){radius=0.0;}
 Circle (double r){radius =r;}
 public double getArea(){
 return radius*radius*3.14;
 }
}

class Rectangle extends GeometricObject{
 private double width, height;
 Rectangle(){width=0.0; height=0.0;}
 Rectangle(double w, double h){width=w; height=h;}
 public double getArea(){
 return width*height;
 }
}

```

*Implement*

Answer the following questions:

- Find out the output of the above program if there is no error. 2
- Rewrite the above java program where illustrate the abstract class 3
- What do you mean by interface in java? Rewrite the above program where declare GeometricObject as an interface instead of a class. 5
- Write down the differences between abstract class and interface. 2

Q.8.

```

package ja04;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.locks.Lock;
import java.util.concurrent.locks.ReentrantLock;

public class JA04{
 private static Math math = new Math();
 public static void main(String[] args){
 ExecutorService executor = Executors.newCachedThreadPool();
 for(int i=0;i<100;i++){
 executor.execute(new Add());
 }
 executor.shutdown();
 while (!executor.isTerminated()){ }
 System.out.println("Result "+math.get());
 }

 private static class Add implements Runnable{
 public void run(){
 math.increment(5);
 }
 }

 private static class Math{
 private int x=0;
 public int get(){
 return x;
 }

 public void increment(int i){
 int newX=x+i;
 try{
 Thread.sleep(5);
 }
 catch (InterruptedException ex){ }
 x=newX;
 }
 }
}

```

}  
}  
}  
Answer the following questions:

(a) If the above program is executed, the following output has been found randomly: 4

Result 5

But coder think, the output must be

Result 500

There is a problem. Your task is to explain this problem according to code.

(b) There are several ways to correct the problem. Give two implicit solutions in this problem and explain your solutions. 4

(c) We know that Lock and conditions can be explicitly used to synchronize threads in java. Rewrite the above program where the problem is solved using Lock interface explicitly. 4

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## RAJSHAHI UNIVERSITY OF ENGINEERING &amp; TECHNOLOGY

## DEPARTMENT OF COMPUTER SCIENCE &amp; ENGINEERING

1<sup>st</sup> Year Even Semester Examination 2017

COURSE NO: CSE 1203 COURSE TITLE: Object Oriented Programming

FULL MARKS: 72

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.  
 (ii) Figures in the right margin indicate full marks.  
 (iii) Use separate answer script for each section.

SECTION : A

- Q1. (a) What is object oriented programming? List the features of object oriented programming. 4  
 (b) Write a program to declare a class "employee" consisting of data members' emp\_no and emp\_name. Write the member functions accept() to accept and display() to display the data for five employees. 4  
 (c) What is a constructor? What is a destructor? When are they executed? 2  
 (d) Find error, if any, from the following code. 2

```
#include<iostream.h>
class xyz{
 private: int d1;
 float d2;
 public: void xyz();
 void display();
};
void main(){
 xyz s;
 s.showdata();
}
```

- Q2. (a) Explain the concept of "this" pointer. 2  
 (b) What happens when a protected member is inherited as public? What happens when it is inherited as private? What happens when it is inherited as protected? 3  
 (c) Given the following base class, 4

```
class area_cl{
 public:
 double height;
 double width;
};
```

Create two derive classes called rectangle and isosceles that inherit area\_cl. Each class includes a function called area() that returns the area of a rectangle or isosceles triangle, as appropriate. Use parameterized constructors to initialize height and width.

- (d) What is runtime polymorphism? Compare overloading and overriding with an example. 3
- Q3. (a) In procedure oriented programming, all data are shared by all functions. Is this statement true? Justify your answer. 2  
 (b) What do you mean by copy constructor? Why do you need to use copy constructor? Explain with an example. 4  
 (c) Write down some attributes of using friend function. 3  
 (d) Why are the following two overloaded functions inherently ambiguous? 3

```
int f(int a);
int f(int &a);
```

- Q4. (a) What are the advantage(s) of call by reference over call by value approach? Discuss the problem(s) and solution(s) of call by reference as well. 5  
 (b) Design a template for calculating the minimum of two objects. Test your template for integer, float and string type objects. 4  
 (c) Define the term: super, casting object and final. 3

SECTION : B

- Q5. (a) What will be the output of the following program? 4
- ```
void execute(int x, int y=200){
    int temp = x+y;
    x+=temp;
    if(y!=200)
        cout<<temp<<x<<y<<endl;
```

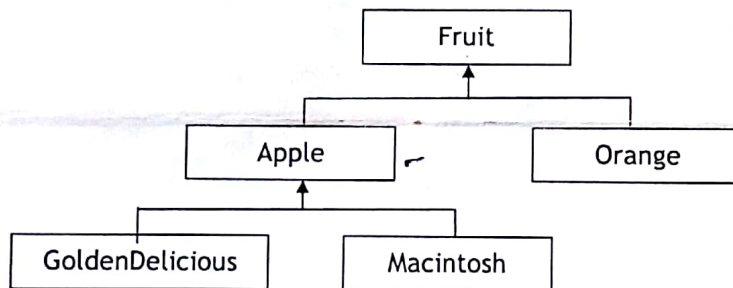
P.T.O

```

void main() {
    int a = 50, b = 20;
    execute(b);
    cout << a << b << endl;
    execute(a, b);
    cout << a << b << endl;
}

```

- (b) Write short note on: 4
- (i) Exception handling mechanism.
 - (ii) Friend function and friend class.
- (c) What will be the status of C++ I/O file if we use the following flags: Write down the meaning of each. (i) goodbit (ii) failbit (iii) badbit (iv) eofbit 4
- Q.6. (a) Differentiate between the following case: (i) int* p; (ii) const int* p; 4
- (iii) const int* const p; (iv) int* const p;
- (b) Discuss about the default constructors that C++ provides. 4
- (c) What is the key principle behind operator overloading? 4
- Q.7. (a) How does java support multiple interfaces? Define how a class or interface inherits an interface. 4
- (b) Write a java program to create two threads, one thread will print odd numbers and second thread will print even numbers between 1 to 20. 4
- (c) What is meant by synchronization? With an example, state why do we need it? 4
- Q.8. (a) How do you explicitly invoke a superclass's constructor from subclass? Explain with an example. 4
- (b) Suppose that, Fruit, Apple, Orange, GoldenDelicious Apple and Macintosh Apple are declared as shown in figure. 5



Assume that the following declaration is given:

```

Fruit fruit = new GoldenDelicious();
Orange orange = new Orange();

```

Answer the following questions:

- i) Is fruit instance of Fruit?
 - ii) Is fruit instance of Orange?
 - iii) Is fruit instance of Apple?
 - iv) Is fruit instance of GoldenDelicious?
 - v) Is fruit instance of Macintosh?
 - vi) Is orange instance of Orange?
 - vii) Is orange instance of Apple?
 - viii) Suppose the method makeApple is defined in the Apple class. Can fruit invoke this method? Can Orange invoke this method?
 - ix) Is the statement Orange P = new Apple() legal?
 - x) Is the statement Macintosh P = new Apple() legal?
 - xi) Is the statement Apple P = new Macintosh() legal?
- (c) What modifier should you use on a class so that a class in the same package can access it, but a class in a different package cannot access it? 3